

Avi Tripathi

Design Engineer | UX and Machine Learning

01 EDUCATION

Sept 2020 — Jun 2024

Imperial College London

MEng Design Engineering

- **Key Modules:** Human-Centered Design, Machine Learning, Enterprise Roll-out, Innovation and Entrepreneurship, UX Design
- **Online Design Portfolio:** avitripathi.com

02 EXPERIENCE

09/2024 — Present

ML Engineer/UX Designer at Autodesk

- Conducted Machine Learning research for the innovation team for fusion manufacturing producing **3 novel customer workflow POC's** for Autodesk Fusion in my first year.
- Integrated foundational and custom encoder models into **full-stack prototypes using figma, react, typescript**, with a novel ML solution released to **15,000+ insider customers**.
- Presented machine learning talks at the Autodesk Developer Conference, to 100s of devs.

09/2023 — 09/2024

ML Engineer/UX Designer at Setmixer (start-up)

- Developed end-to-end **Figma UX/UI Mockups** for music streaming website, desktop and mobile. Supported in development of React/Typescript app, now released to users - <https://listen.setmixer.com/>
- Led development of advanced live musical instrument classification ML model, achieving **3% performance improvement over state-of-the-art**, using CNNs and transfer learning.

09/2023 — 07/2024

Founding XD Engineer at Patent Space (start-up)

- Built an infringement risk management start-up from scratch which was a **finalist of the London Mayor's Entrepreneur Competition 2024**.
- Led development of **UX and front end prototypes using Figma**, for an **AI enriched patent search and infringement monitoring tool**, backed by user research and interviews.

08/2022 — 08/2024

Brand Manager at ICL Design Engineering Society

- Maintained the brand for a society of **over 500** students through design of media and merchandise, whilst also leading advertising and supply, achieving **over £3000** in sales.
- Individually created the visual identity for **2 Open House events** held with the Great Exhibition Road Festival, attracting over **50,000 visitors**.

03 KEY SKILLS

XD | Figma (3 years)

Graphics | Ps, Ai (5 years)

Motion | Ae, Blender (3 years)

ML Models | LLMs, Encoders, Pytorch

WebDev | React, Typescript

Scripting | Python, C++